Research in Developmental Disabilities 175 (2026) 105342



Contents lists available at ScienceDirect

Research in Developmental Disabilities

journal homepage: www.elsevier.com/locate/redevdis





A systematic review on the association between ADHD and procrastination

Rossella Suriano *

Department of Cognitive Sciences, University of Messina, Messina 98100, Italy

ARTICLE INFO

Keywords:
ADHD
Procrastination
Self-regulation
Systematic review
PRISMA

ABSTRACT

Background: Procrastination is a maladaptive self-regulation behavior frequently observed in individuals with attention-deficit/hyperactivity disorder (ADHD), yet the consistency and underlying mechanisms of this association remain unclear.
Methods: Following PRISMA 2020 guidelines, a systematic search was conducted in Web of Science, Scopus, PsycINFO, and PubMed. Peer-reviewed quantitative studies examining the relationship between ADHD (diagnosis or symptoms) and procrastination using validated measures were included.
Results: Eleven studies (N = 2788) met inclusion criteria. Across studies, ADHD symptoms were consistently positively associated with procrastination. Inattention emerged as the most robust correlate. Findings were also consistent with the involvement of executive functioning deficits, emotion dysregulation, and motivational processes, although evidence was primarily based on cross-sectional designs.
Conclusion: The available evidence supports a robust association between ADHD symptoms and procrastination. However, methodological limitations, including reliance on cross-sectional data and heterogeneous measures, preclude causal inferences. Future research should employ longitudinal and multi-method designs to clarify underlying mechanisms.

1. Introduction

Procrastination is commonly defined as the voluntary and irrational delay in initiating or completing a planned action, despite awareness of its potential negative consequences ((Klingsieck, 2013a, 2013b; Steel, 2007; Svartdal & Nemtcan, 2022). The construct has been progressively reconceptualized as multidimensional, encompassing different forms (e.g., general, academic, decisional, and behavioral) and distinguishing between functional and dysfunctional configurations (active vs. passive procrastination) (Choi & Moran, (2009); Chun Chu & Choi, (2005); Ferrari, (1992); (Klingsieck, 2013a, 2013b). This theoretical evolution has helped move beyond a unitary view of the phenomenon, instead framing it as the expression of dysfunctional processes of cognitive, emotional, and motivational regulation (Rebetez et al., 2015; Sirois & Pychyl, 2013).

More recent theoretical models converge in interpreting procrastination as a failure of self-regulation, arising from the interaction between deficits in executive functions, difficulties in emotion regulation, and alterations in motivational processes (Mohammadi Bytamar et al., 2020; Rebetez et al., 2018; Steel, 2007; Zhang et al., 2019). In particular, the inability to maintain focus on long-term

* Correspondence to: Department of Cognitive, Psychological and Pedagogical Sciences and Cultural Studies, University of Messina, Messina 98100, Italy.
E-mail address: rossella.suriano@studenti.unime.it.

https://doi.org/10.1016/j.ridd.2026.105342
Received 14 April 2026; Received in revised form 28 June 2026; Accepted 13 July 2026
Available online 15 July 2026
0891-4222/© 2026 The Author. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

goals, modulate negative emotional states associated with the task, and resist immediate gratification represents a core feature of the phenomenon (Wu et al., 2016). These perspectives are situated within the framework of Temporal Motivation Theory (Steel & König, 2006) and broader models of self-regulatory failure, which highlight the role of impulsivity, delay sensitivity (delay discounting), and emotion regulation processes in the development of procrastinatory behaviors (Steel et al., 2018).

Consistent with dual-process models of self-regulation (Hofmann, Friese, & Strack, 2009), procrastination can be further interpreted as the result of an imbalance between controlled (top-down) processes, responsible for maintaining goal-directed behavior, and automatic (bottom-up) processes, driven by impulses, emotional reactivity, and sensitivity to immediate reward. From this perspective, procrastinatory behaviors emerge when automatic processes—such as avoidance tendencies triggered by negative emotions associated with the task or a preference for immediate gratification—override executive control mechanisms (Sirois, 2014; Zentall, 2021). Accordingly, procrastination may be understood as a short-term regulatory strategy aimed at reducing immediate distress, although dysfunctional with respect to the pursuit of future goals (Sirois & Pychyl, 2013).

Within this theoretical framework, conditions characterized by impairments in self-regulatory processes are particularly relevant. Among these, attention-deficit/hyperactivity disorder (ADHD) represents a key reference model (Bolden & Fillauer, 2020; Oguchi et al., 2021). ADHD is a neurodevelopmental disorder characterized by persistent and pervasive patterns of inattention, hyperactivity, and impulsivity, and is classified into three clinical presentations: predominantly inattentive (ADHD-I), predominantly hyperactive/impulsive (ADHD-H), and combined (ADHD-C) (American and Association, 2013; Fabio et al., 2024; Faraone et al., 2021). Within recent theoretical models, ADHD has been conceptualized as a dual-pathway disorder (Sonuga-Barke, 2002), in which deficits in executive control processes coexist with alterations in motivational systems, including delay aversion and heightened sensitivity to reward (Karalunas & Huang-Pollock, 2011; Sonuga-Barke, 2003). Beyond this framework, broader accounts have suggested that individuals with ADHD may also exhibit steeper temporal discounting, reflecting a stronger preference for immediate over delayed outcomes (Marx et al., 2021). These motivational characteristics may contribute to difficulties in sustaining effort toward long-term goals and help explain the association between ADHD symptoms and procrastinatory behavior.

Individuals with ADHD exhibit significant impairments across multiple domains, including academic, social, emotional, and physical functioning, with substantial consequences for everyday life (French et al., 2024; Kosheleff et al., 2023). A large body of literature has documented a robust association between the core symptoms of the disorder and deficits in executive functions, including working memory, inhibitory control, cognitive flexibility, and planning ability (Fabio & Suriano, 2021; 2024; Martinussen et al., 2005; Willcutt et al., 2005). These difficulties, together with alterations in motivational and emotional processes (e.g., delay aversion and difficulties in affective regulation), support the conceptualization of ADHD as a disorder of executive and behavioral regulation, as well as a condition characterized by an imbalance between top-down control systems and bottom-up motivational and automatic processes (Fabio et al., 2026; Petrovic & Castellanos, 2016; Sonuga-Barke, 2002).

From this perspective, the mechanisms underlying ADHD show substantial overlap with those implicated in procrastination, suggesting the involvement of transdiagnostic self-regulatory processes in the link between the two constructs (Bodalski et al., 2023; Niermann & Scheres, 2014). Consistently, a growing number of empirical studies have investigated the association between ADHD symptoms and procrastinatory behaviors in both clinical and non-clinical populations (Altgassen et al., 2019; Niermann & Scheres, 2014). However, the existing literature remains fragmented and is characterized by marked heterogeneity in the definition and operationalization of both ADHD (e.g., symptoms vs. diagnosis) and procrastination (e.g., global vs. domain-specific measures). In this context, distinguishing among different forms of procrastination is particularly important, because global, academic, decisional, behavioral, active, and passive procrastination may reflect partially distinct self-regulatory processes and may therefore be associated with different patterns of ADHD symptoms. Previous research has used heterogeneous measures of procrastination and has often focused on specific manifestations of the construct, leaving open the question of whether the ADHD–procrastination association is comparable across different forms of procrastinatory behavior. Moreover, the principal shared theoretical mechanisms—such as emotion regulation, delay sensitivity, and the imbalance between automatic and controlled processes—have rarely been considered in an integrated manner.

In this regard, a systematic synthesis that critically organizes the empirical evidence within a unified theoretical framework appears crucial for advancing understanding of the phenomenon and clarifying the nature of the relationship between ADHD and procrastination. Despite the growing body of empirical research, findings remain fragmented, and no systematic synthesis has yet clarified whether the association between ADHD and procrastination is consistent across different populations and operationalizations, nor which mechanisms most reliably account for this relationship.

In light of this, the present systematic review aims to examine the quantitative empirical literature on the relationship between ADHD symptoms (or diagnosis) and procrastination. Specifically, the objectives of the review are to: (a) examine the consistency of the association between ADHD and procrastination across studies; (b) analyze the role of different forms of procrastination; (c) explore the principal cognitive, emotional, and motivational mechanisms underlying this relationship; and (d) identify the main methodological and theoretical gaps in the existing literature.

2. Method

2.1. Search strategy

The systematic literature search was conducted across four electronic databases: Scopus, Web of Science, PsycINFO, and PubMed. The last search was conducted on March 15, 2026. The general search string was: (ADHD OR "attention deficit" OR inattention OR impulsiv* OR hyperactiv*) AND procrastinat*. The search strategy was adapted to the syntax and indexing systems specific to each

database in order to ensure conceptual equivalence across sources (e.g., title/abstract fields in PubMed and PsycINFO, topic search in Web of Science, and title–abstract–keywords fields in Scopus). The complete database-specific search strategies, including field codes and applied filters, are reported in Supplementary Table S1. Searches were limited to peer-reviewed scientific articles published in English, with no temporal restrictions. No additional hand-searching of reference lists was conducted.

2.2. Eligibility criteria

Studies were included in the present systematic review if they met the following criteria.

2.3. Inclusion criteria

Only empirical quantitative studies were included, encompassing both cross-sectional and longitudinal designs. Studies were required to assess attention-deficit/hyperactivity disorder (ADHD) through either a formal clinical diagnosis or validated psychometric instruments specifically designed to measure ADHD symptoms (e.g., ASRS, CAARS, Conners scales), operationalized either categorically or dimensionally.

Procrastination had to be measured using standardized and validated instruments (e.g., General Procrastination Scale, Procrastination Assessment Scale for Students, Pure Procrastination Scale), ensuring construct validity and comparability across studies.

Studies were included only if they explicitly and directly examined and statistically tested the relationship between ADHD and procrastination within the main analyses. Eligible analytical approaches included correlations, regression analyses, group comparisons, structural equation modeling, and mediation or moderation analyses, provided that the ADHD–procrastination association was a primary or central component of the main analytical framework. Studies including additional psychological or behavioral variables were retained, provided that the ADHD–procrastination association was explicitly tested within this framework.

The reference to a "central role" was not intended to introduce an additional eligibility criterion, but rather to clarify that the ADHD–procrastination association had to be part of the study's main analytical framework. Specifically, the association was considered central when procrastination was operationalized as a primary outcome, mediator, moderator, or key construct analyzed in relation to ADHD within the main statistical analyses, rather than merely as a descriptive, secondary, or peripheral variable.

2.4. Exclusion criteria

Studies were excluded if they met one or more of the following conditions.

Studies that did not explicitly examine or statistically test the relationship between ADHD and procrastination were excluded, even when both constructs were measured.

Studies were also excluded when ADHD or procrastination was included only as a covariate, control variable, descriptive variable, secondary variable, or marginal predictor in analyses primarily focused on other relationships, without a direct statistical test of the ADHD–procrastination association.

Studies were further excluded when ADHD was used only as a moderator of the relationship between procrastination and another outcome, or when procrastination was used only as a predictor, covariate, or associated factor of another outcome, without the ADHD–procrastination association itself being a primary or central component of the main analysis.

Studies using non-validated, non-specific, or ad hoc measures of ADHD or procrastination were excluded. This restriction was adopted to preserve the conceptual specificity of the review and to ensure that the included evidence directly addressed the ADHD–procrastination association, rather than only providing ancillary or descriptive information from analyses primarily focused on other constructs.

2.5. Selection procedure

The screening of studies was conducted using Microsoft Excel. In an initial phase, the identified records were examined on the basis of title and abstract in order to assess their relevance with respect to the inclusion and exclusion criteria. Subsequently, articles deemed potentially eligible were evaluated at the full-text level. During the screening process, studies were classified as included, excluded or potentially relevant based on the criteria defined previously. Two independent reviewers screened titles and abstracts based on the predefined eligibility criteria. Full-text articles were then assessed independently. Disagreements were resolved through discussion, and when necessary, consultation with a third experienced researcher. Studies that measured both ADHD-related variables and procrastination but did not directly test their association were excluded. Study authors were not contacted to obtain or confirm unreported associations.

2.6. Data extraction

For each included study, data were extracted using a standardized extraction form. The extracted information included: authors, year of publication, country, sample characteristics, study design, measurement instruments for ADHD and procrastination, and the main findings.

2.7. Quality assessment

The methodological quality of the included studies was independently assessed by two reviewers using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Analytical Cross-Sectional Studies. The checklist evaluates eight domains related to sampling procedures, measurement validity, confounding factors, and statistical analysis. Each item was rated as "yes", "no", "unclear", or "not applicable".

2.8. Data synthesis

Due to substantial heterogeneity in study designs, measures, and statistical approaches, a meta-analysis was not conducted. Specifically, the included studies differed in the operationalization of ADHD, including clinical diagnosis, screening cut-offs, dimensional symptom scores, and specific symptom dimensions. They also differed in the type of procrastination assessed, including general, academic, passive, active, and behavioral procrastination, as well as in sample characteristics, ranging from clinical adult samples to university, medical student, community, and online samples. In addition, the statistical indices reported across studies were not directly comparable, as they included zero-order and partial correlations, standardized and unstandardized regression coefficients, group comparisons, mediation paths, SEM estimates, and indirect effects. Several studies also did not report confidence intervals or sufficient information to transform all associations into a common standardized effect size without introducing additional assumptions. For these reasons, a formal meta-analysis was not considered appropriate. Instead, a narrative synthesis was performed, while tabulating the main effect estimates and precision information when available. The synthesis focused on (1) the consistency of associations between ADHD symptoms and procrastination and (2) the explanatory mechanisms proposed across studies, including executive functioning, emotion regulation, and motivational processes. Particular attention was given to differences in sample characteristics, ADHD operationalization, types of procrastination assessed, and study quality. Studies were grouped narratively according to sample characteristics, ADHD operationalization, type of procrastination assessed, and the explanatory mechanisms examined. The review was not prospectively registered, and no protocol was prepared.

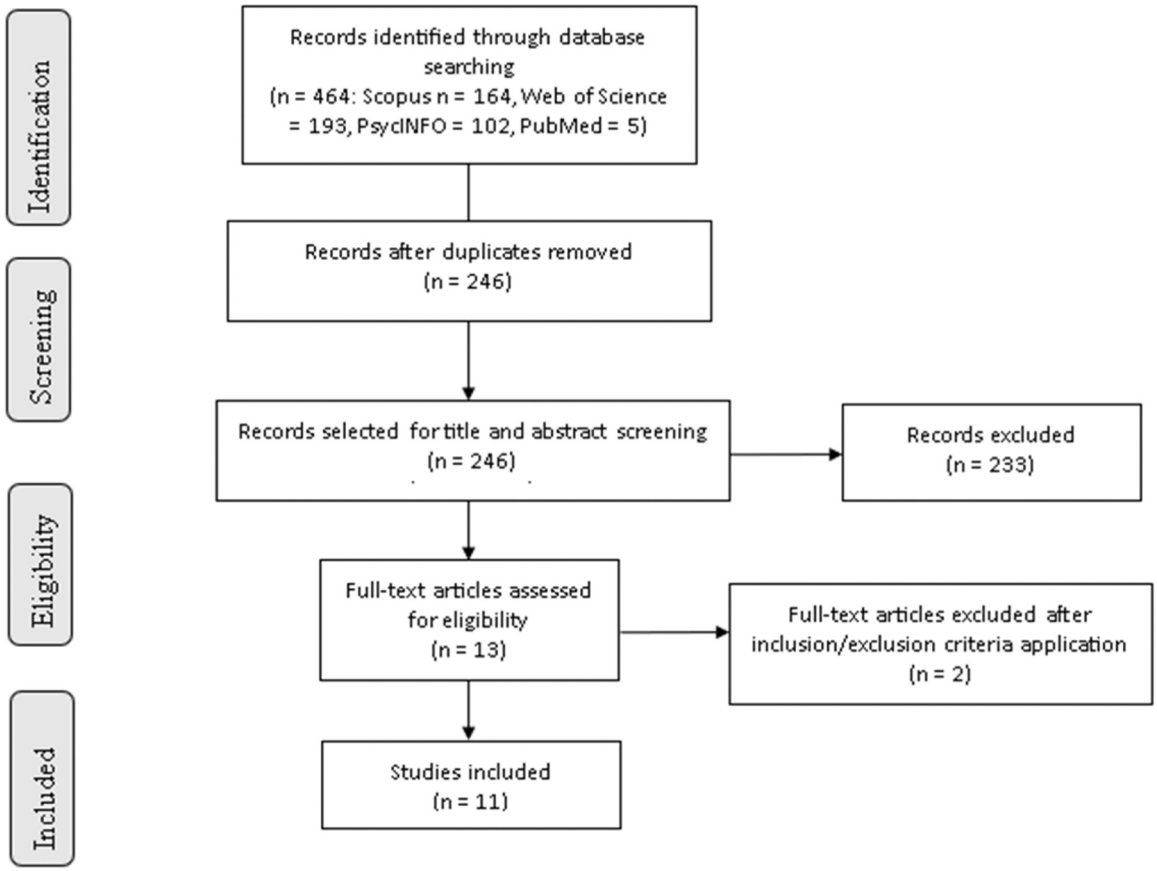


Fig. 1. PRISMA diagram for study selection.

Table 1
Characteristics of studies.

Authors (year)	Country/ recruitment context	Sample (N)	Age (M ± SD / Range)	Type of sample	ADHD operationalization	Study design	Main variables analyzed	ADHD–procrastination statistical association	Main findings
Altgassen et al. (2019)	Germany	N = 53 (29 ADHD, 24 controls)	~36–38 years	Clinical adult sample	Clinical DSM−5 diagnosis + symptom questionnaire (FEA-ASB)	Cross-sectional (mediation analysis)	Procrastination (Tuckman), prospective memory (lab + real-life), future orientation	ADHD–procrastination: b = .76, p < .001; mediation via real-life prospective memory: Sobel z = 2.31, p = .02; 95% CI not reported.	Adults with ADHD reported greater procrastination, partly explained by poorer everyday prospective memory.
Ashworth & McCown (2018)	USA	N = 84	M = 35.39 ± 10.32	Clinical adult sample (outpatients evaluated for ADHD)	ADHD symptoms via ASRS	Cross-sectional	Procrastination, ADHD symptoms, attention performance (auditory & visual), hoarding	ADHD–procrastination: r = .424, p < .001, 95% CI [.227,.580]; after removing the overlapping ASRS item: r = .401, p < .001.	Procrastination was positively associated with ADHD symptoms, even after accounting for item overlap.
Bodalski et al. (2023)	USA	N = 595	Range: 18–26 + (94.3% between 18–22)	University students (non-clinical)	ADHD symptoms via DSM−5 ADHD self-report scale	Cross-sectional (mediation & serial mediation)	Procrastination, emotion dysregulation, self-esteem	ADHD–procrastination: r = .423, p < .001; mediation via emotion dysregulation: ab = .24, 95% CI [.17,.32]; mediation via self-esteem: ab = .16, 95% CI [.102,.230]; serial mediation: ab = .10, 95% CI [.05,.15].	Emotion dysregulation and self-esteem partly explained the ADHD–procrastination association.
Bolden & Fillauer (2020)	USA	N = 114	M = 18.79 ± 1.13	University students (non-clinical)	ADHD symptoms via BAARS-IV (self-report + interview)	Cross-sectional (mediation analysis)	Procrastination, ADHD symptoms (inattention, hyperactivity, impulsivity, SCT), executive functions	ADHD–procrastination: r = .462, p < .001; mediation via executive functions: Sobel z = 5.81, p < .001; indirect effects via time management: ab = .46, 95% CI [.33,.62], and organization/ problem-solving: ab = .30, 95% CI [.21,.42].	Executive-function difficulties, especially time management and organization/problem-solving, explained the association between procrastination and ADHD symptoms.
Massoodi et al. (2025)	Iran	N = 256	M = 24.30 ± 1.83	Medical students	ADHD symptoms via ASRS	Cross-sectional (correlational & regression)	Academic procrastination, academic burnout	ADHD–procrastination: r = .478, p < .001; ASRS predicting procrastination: β = .746, 95% CI [.569,.924], p < .001.	Higher ADHD symptoms were associated with greater academic procrastination.
Müller et al. (2023)	Hungary	N = 408	M = 23.37 ± 3.87	University students	ADHD symptoms via ASRS (cut-off screening)	Cross-sectional	ADHD symptoms, procrastination (active/passive), depression, boredom, smartphone addiction	ADHD–procrastination: passive procrastination, β = .31, p < .001; active procrastination, β = −.12, p = .002; model R^2adj = .41; 95% CI not reported.	Passive procrastination was positively associated with ADHD symptoms, whereas active procrastination showed a negative association.
Netzer Turgeman & Pollak (2023)	Israel	N = 202	M = 26.23 ± 7.10 (18–50)	Students and general population	ADHD symptoms via ASRS (dimensional, item overlap removed)	Cross-sectional (mediation study)	Procrastination (IPS), expectancy, task aversiveness, impulsiveness	ADHD–procrastination: B = .194, 95% CI [.031,.356]; indirect effects via expectancy: B = .056, 95% CI [.016,.106], and impulsiveness: B = .276, 95% CI [.128,.428].	Lower expectancy and higher impulsiveness partly explained the ADHD–procrastination association.
Netzer Turgeman & Pollak (2025)	Online MTurk sample	N = 132	M = 39.9 ± 9.3 (24–60)	Community adult sample	ADHD symptoms via ASRS (dimensional)	Cross-sectional (mediation)	Procrastination (IPS), quality of life (AAQoL, WHOQOL-BREF)	ADHD–procrastination: r = .72, p < .01; indirect effect via procrastination on AAQoL: B = −0.21, 95% CI [−0.38, −0.07].	Procrastination partly explained the association between ADHD symptoms and poorer quality of life.

(*continued on next page*)

Table 1 (*continued*)

Authors (year)	Country/ recruitment context	Sample (N)	Age (M ± SD / Range)	Type of sample	ADHD operationalization	Study design	Main variables analyzed	ADHD–procrastination statistical association	Main findings
Netzer Turgeman & Pollak (2026)	Online Prolific sample	N = 640	M = 36 ± 9.64 (18–56)	General adult sample (online)	ADHD symptoms via ASRS (inattention dimension)	Cross-sectional (SEM, mediation model)	Procrastination	ADHD–procrastination: total effect B = 1.061, 95% CI [.964, 1.174]; direct effect B = .465, 95% CI [.259,.650]; total indirect effect B = .595, 95% CI [.435,.803].	Low task value, sensitivity to delay, and flexible goal adjustment partly explained the ADHD inattention–procrastination association.
Niermann & Scheres (2014)	Netherlands	N = 54	M = 19.72 ± 2.05	Undergraduate students	ADHD symptoms (inattention, hyperactivity, impulsivity – self-report)	Cross-sectional (correlational + behavioral task)	Procrastination (general, academic, behavioral)	ADHD–procrastination: inattention correlated with the Pure Procrastination Scale, $r = .51$, $p < .01$; partial $r = .43$, $p < .01$. Hyperactivity/impulsivity was not significant after controlling for inattention.	Inattention, but not hyperactivity/ impulsivity, was specifically associated with general procrastination.
Shirdel et al. (2024)	Iran	N = 250	M = 27.27 ± 8.84	Community adult sample	ADHD symptoms via ASRS	Cross-sectional (SEM, mediation)	Procrastination, ADHD symptoms, SCT, driving behavior	ADHD–procrastination: $\beta = .52$, $p < .05$; indirect effect via procrastination on driving behavior: $\beta = .19$, $p < .05$; 95% CI not reported.	Procrastination mediated the association between ADHD symptoms and problematic driving behavior.

3. Results

3.1. Study selection

The initial search in the databases yielded a total of 464 records. After the removal of 218 duplicates, 246 articles remained and were subjected to screening based on title and abstract. Subsequently, articles deemed potentially relevant (13) were evaluated through full-text review according to the inclusion and exclusion criteria previously defined. Of the 13 articles assessed for eligibility at the full-text stage, 2 were excluded. Reasons for exclusion are provided in Supplementary Table S2. At the end of this selection process, 11 studies were included in the present systematic review (Fig. 1).

3.2. Characteristics of the included studies

The included studies, published between 2014 and 2026, comprised a total of 2788 participants from studies conducted across six countries: Germany, the United States, Iran, Hungary, Israel, Netherlands, and online samples recruited through MTurk or Prolific. Sample sizes ranged from 53 participants (Altgassen et al., 2019) to 640 participants (Netzer Turgeman & Pollak, 2026).

Most studies were conducted on non-clinical samples, including university students, medical students, and general adult online samples (Bodalski et al., 2023; Bolden & Fillauer, 2020; Massoodi et al., 2025; Müller et al., 2023; Netzer Turgeman & Pollak, 2023, 2025, 2026; Niermann & Scheres, 2014; Shirdel et al., 2024). A more limited number included clinical or clinically characterized samples, such as adults with a clinical ADHD diagnosis or outpatients evaluated for ADHD (Altgassen et al., 2019; Ashworth & McCown, 2018).

All studies adopted a cross-sectional design; several studies additionally employed mediation models or structural equation models (Bodalski et al., 2023; Bolden & Fillauer, 2020; Netzer Turgeman & Pollak, 2023, 2026; Shirdel et al., 2024).

ADHD was operationalized both categorically (clinical diagnosis; Altgassen et al., 2019) and dimensionally using validated self-report instruments, predominantly the Adult ADHD Self-Report Scale (ASRS) (Ashworth & McCown, 2018; Massoodi et al., 2025; Müller et al., 2023; Netzer Turgeman & Pollak, 2023, 2025, 2026; Shirdel et al., 2024). Procrastination was assessed in all studies using standardized scales, with variations in type (general procrastination, academic procrastination, or specific subtypes). The main characteristics of the included studies are reported in Table 1.

3.3. Association between ADHD and procrastination

All included studies reported at least one significant positive association between ADHD symptoms and procrastination, although the magnitude of the effects and the operationalization of both constructs varied across studies (Altgassen et al., 2019; Ashworth & McCown, 2018; Bodalski et al., 2023; Bolden & Fillauer, 2020; Massoodi et al., 2025; Müller et al., 2023; Netzer Turgeman & Pollak, 2023, 2025, 2026; Niermann & Scheres, 2014; Shirdel et al., 2024).

Although the heterogeneity of statistical metrics precluded quantitative pooling, the available estimates suggest that the most frequent pattern was a moderate positive association between ADHD symptoms and procrastination. Among studies reporting zero-order correlations, effect sizes generally fell in the moderate range, approximately between $r = .42$ and $r = .51$, with one study reporting a stronger association ($r = .72$). Regression and SEM-based estimates were less directly comparable, but they generally converged in indicating a positive association between ADHD symptoms and procrastination. For example, passive procrastination showed a positive association with ADHD symptoms, whereas active procrastination showed a small negative association, suggesting that the magnitude and direction of effects may vary depending on the specific form of procrastination assessed. Overall, the distribution of findings therefore points to a predominantly moderate positive association, while also indicating meaningful variability across measures and analytical models.

In clinical samples, individuals with ADHD showed significantly higher levels of procrastination compared to controls (Altgassen

Table 2
Methodological Quality Assessment (JBI Checklist for Analytical Cross-Sectional Studies).

Study	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Altgassen et al., (2019)	Y	Y	Y	Y	Y	Y	Y	Y
Ashworth & McCown, (2018)	Y	Y	Y	Y	N	N	Y	U
Bodalski et al., (2023)	Y	Y	Y	Y	Y	Y	Y	Y
Bolden & Fillauer, (2020)	Y	Y	Y	Y	Y	Y	Y	Y
Massoodi et al., (2025)	Y	Y	Y	Y	U	U	Y	Y
Müller et al., (2023)	Y	Y	Y	Y	Y	Y	Y	Y
Netzer Turgeman & Pollak, (2023)	Y	Y	Y	Y	Y	Y	Y	Y
Netzer Turgeman & Pollak, (2025)	Y	Y	Y	Y	Y	Y	Y	Y
Netzer Turgeman & Pollak, (2026)	Y	Y	Y	Y	Y	Y	Y	Y
Niermann & Scheres, (2014)	Y	Y	Y	Y	U	Y	Y	Y
Shirdel et al., (2024)	Y	Y	Y	Y	Y	Y	Y	Y

Note: Q1 = inclusion criteria clearly defined; Q2 = study subjects and setting described in detail; Q3 = exposure measured in a valid and reliable way; Q4 = objective or standard criteria used for measurement of the condition; Q5 = confounding factors identified; Q6 = strategies to address confounding factors reported; Q7 = outcomes measured in a valid and reliable way; Q8 = appropriate statistical analysis used.

et al., 2019), whereas in symptom-based samples, ADHD symptom severity was consistently associated with higher procrastination (Ashworth & McCown, 2018; Massoodi et al., 2025). For instance, Massoodi et al. (2025) reported a moderate correlation between ADHD and academic procrastination ($r = .478$), along with a significant predictive effect ($\beta = .746$).

This association was also observed in non-clinical populations. Among university students, ADHD symptoms were positively associated with procrastination in both correlational analyses and mediation models (Bodalski et al., 2023; Bolden & Fillauer, 2020). Notably, in the latter study, executive functions statistically mediated the relationship in a cross-sectional model in which procrastination was treated as the predictor and ADHD symptoms as the outcome, limiting conclusions about directional effects. Similarly, Müller et al. (2023) found that passive procrastination was positively associated with ADHD symptoms, whereas active procrastination showed a negative association.

Converging evidence also emerged from general adult samples, in which ADHD symptoms were positively associated with procrastination even after controlling for motivational variables and item overlap (Netzer Turgeman & Pollak, 2023, 2025, 2026; Shirdel et al., 2024).

Regarding symptom dimensions, several studies indicated that inattention represents the most robust predictor of procrastination. Specifically, inattention was significantly associated with general procrastination, whereas hyperactivity and impulsivity were no longer significant after controlling for shared variance (Niermann & Scheres, 2014). This pattern was supported by other studies, which also found stronger associations between procrastination and inattentive symptoms compared to other ADHD dimensions (Bolden & Fillauer, 2020; Netzer Turgeman & Pollak, 2023, 2026).

3.4. Explanatory Mechanisms and Functional Correlates

Several studies have examined the possible mechanisms underlying the association between ADHD and procrastination, suggesting a multifactorial model involving executive, emotional, and motivational processes.

With regard to executive functions, the findings are consistent with their possible role in the link between ADHD and procrastination. In particular, Bolden and Fillauer (2020) reported evidence compatible with a mediating role of executive functions, with particular relevance for time management and organization. Consistent findings also emerge from clinical studies, in which poorer attentional performance was associated with greater procrastination (Ashworth & McCown, 2018). In addition, Altgassen et al. (2019) reported findings compatible with a partially mediating role of prospective memory in the relationship between ADHD and procrastination.

Emotional processes also emerged as relevant. Bodalski et al. (2023) reported findings consistent with a role of emotion dysregulation and low self-esteem in the relationship between ADHD symptoms and procrastination, including a significant serial pathway.

At the motivational level, Netzer Turgeman and Pollak (2023) found that lower expectancy of success and greater impulsivity were associated with the relationship between ADHD and procrastination, with results compatible with a mediating role. In a subsequent study, the same authors (2026) further highlighted the possible involvement of delay sensitivity, low task value, and flexible goal regulation.

Finally, some studies examined procrastination as a possible intermediate process between ADHD and functioning. In particular, the findings are consistent with a role of procrastination in the relationship between ADHD and quality of life (Netzer Turgeman & Pollak, 2025), as well as between ADHD and dysfunctional driving behaviors (Shirdel et al., 2024).

Overall, these findings are consistent with the hypothesis that procrastination may represent an important intermediate process through which ADHD symptoms are associated with functional difficulties in everyday life; however, the cross-sectional nature of the included studies does not allow causal conclusions to be drawn.

3.5. Quality assessment

The methodological quality of the included studies is presented in Table 2. Overall, the studies demonstrated generally sound methodological characteristics across most domains. All studies clearly defined their inclusion criteria (Q1) and provided adequate descriptions of participants and study settings (Q2). Exposure and outcome variables were assessed using validated and reliable instruments in the majority of studies (Q3, Q7), and appropriate statistical analyses were consistently employed (Q8).

However, some limitations were identified in relation to confounding. Although several studies incorporated multivariate statistical approaches, including regression, mediation, or structural equation modeling (Q6), the explicit identification and conceptual discussion of potential confounding factors (Q5) were not consistently reported across all studies.

Overall, these findings indicate variability in specific methodological domains, particularly concerning the identification and handling of confounding variables, which should be considered when interpreting the results of the included studies.

4. Discussion

The present systematic review provides converging evidence of a positive and consistent association between ADHD symptoms and procrastination, observed across different populations, contexts, and modes of construct operationalization. This pattern emerges in both clinical and non-clinical samples, suggesting that procrastination represents a relatively stable correlate of the difficulties associated with ADHD, rather than a phenomenon limited to specific contextual conditions. In particular, the inattention dimension emerges as the most robust and consistent correlate of procrastination, whereas the contribution of hyperactivity and impulsivity appears less stable when the shared variance among symptom dimensions is taken into account (Bolden & Fillauer, 2020; Niermann &

Scheres, 2014). This finding is consistent with the hypothesis that procrastination is more strongly linked to difficulties in sustained attention, planning, and the regulation of goal-directed behavior (Barkley,1997; Willcutt et al., 2005).

Overall, the findings are consistent with models that interpret procrastination as an expression of self-regulatory difficulties. The included studies show that ADHD symptoms are associated with patterns characterized by reduced goal-directed control, including difficulties in task initiation, time management, and persistence (Bolden & Fillauer, 2020). At the same time, motivational and emotional variables—such as lower expectancy of success, greater impulsivity, heightened delay sensitivity, stronger preference for immediate rewards, and steeper temporal discounting—may help explain the association between ADHD symptoms and procrastination. Rather than pointing to a single explanatory mechanism, the evidence suggests that the relationship between ADHD and procrastination is better understood as the result of the interaction among cognitive, emotional, and motivational processes (Netzer Turgeman & Pollak, 2023; Steel & König, 2006).

More specifically, several studies reported findings compatible with a possible role of executive functions in this association, particularly with regard to planning, organization, and time management (Altgassen et al., 2019). Similarly, the findings support the involvement of emotion regulation processes, highlighting how procrastination is associated with greater emotional distress and lower self-esteem in individuals with higher levels of ADHD symptoms (Bodalski et al., 2023; Mohammadi Bytamar et al., 2020; Sirois & Pychyl, 2013). Motivational variables, including delay aversion and task valuation, also emerge as relevant correlates (Steel et al., 2018). However, it is important to emphasize that these findings should be interpreted as converging associative patterns rather than evidence of specific causal mechanisms, in light of the cross-sectional nature of the included studies. Taken together, these findings are consistent with a multi-component account in which executive dysfunction, emotion regulation difficulties, and motivational alterations jointly contribute to procrastinatory behavior in individuals with higher ADHD symptoms.

A further element that emerged concerns the possible role of procrastination as an intermediate process between ADHD symptoms and functional outcomes. Some studies have reported associations between procrastination and indicators of functioning, such as quality of life and everyday functioning (Kosheleff et al., 2023). Although these findings are consistent with the hypothesis that procrastination may represent a relevant pathway through which ADHD symptoms are associated with functional difficulties, such interpretations should be considered with caution and cannot be taken as evidence of mediation in a causal sense.

At the same time, several important methodological limitations must be considered. First, all included studies adopted cross-sectional designs, which do not allow conclusions to be drawn regarding the directionality of the relationship. It is plausible that this association reflects bidirectional or reciprocal processes, in which ADHD symptoms and procrastination influence each other over time. Moreover, although several studies employed mediation models, these analyses were based on cross-sectional data and therefore cannot be interpreted as evidence of temporal or causal mediation. Second, the predominant use of self-report measures for both ADHD and procrastination introduces the risk of common method variance and subjective reporting biases. In addition, some widely used ADHD symptom scales include items that directly refer to delaying or avoiding cognitively demanding tasks (e.g., difficulty getting started on tasks requiring sustained mental effort), creating potential construct overlap with procrastination measures. Consequently, part of the observed association may reflect shared item content rather than entirely distinct underlying psychological processes. This issue should be considered when interpreting the strength of the reported relationships and highlights the importance of employing multi-method assessment strategies in future research. However, it should also be acknowledged that procrastination is not fully transparent to external observation. By definition, procrastination involves a voluntary delay despite awareness of potential negative consequences, and therefore depends partly on the individual's subjective judgment, intention, and awareness. For this reason, external informant ratings and behavioral measures may complement self-report instruments, but they cannot fully replace them. Experimental or behavioral tasks may capture delay-related behavior, but they may not always distinguish procrastination from other processes, such as forgetting, poor prospective memory, task misunderstanding, or lack of opportunity. Future research should therefore combine self-report, informant-based, and behavioral indicators while carefully distinguishing intentional delay from other forms of task non-completion. In addition, the role of confounding variables has not been systematically addressed. Factors such as depression, anxiety, and personality traits—particularly low conscientiousness—are associated with both ADHD and procrastination and may contribute to the observed shared variance.

Additional limitations concern the predominance of non-clinical samples, particularly university students, which may limit external validity and the generalizability of the findings to clinically diagnosed populations. Furthermore, although the included studies were conducted across multiple countries, the available evidence is derived predominantly from Western, highly educated, or university student samples. Consequently, the extent to which the observed associations generalize across different cultural, socio-economic, and educational contexts remains uncertain, highlighting the need to examine the ADHD–procrastination relationship in more diverse and culturally representative populations. Another important limitation concerns the heterogeneity of procrastination assessment. Although all included studies employed validated instruments, these questionnaires capture partially different facets of the construct, including general procrastination, academic procrastination, and active or passive forms of procrastination. Such conceptual and psychometric differences may contribute to variability in the reported associations with ADHD symptoms and complicate direct comparisons across studies. Future research would benefit from greater consistency in measurement practices and from explicitly distinguishing among different forms of procrastination. Longitudinal designs, the use of multi-method assessments (e.g., behavioral measures and external informant ratings), and more systematic control of confounding factors will also be necessary to clarify the nature of the relationship.

A further limitation concerns the relatively restrictive eligibility criteria adopted in the present review. We focused on studies in which the ADHD–procrastination association was examined as a main objective or as a central component of the analytical framework. This decision was made to preserve the conceptual focus of the review and to ensure that the included studies provided directly interpretable evidence regarding the association of interest. However, this criterion may have reduced the comprehensiveness of the

synthesis, as studies in which ADHD or procrastination was included only as a moderator, covariate, descriptive variable, secondary variable, or ancillary correlate may still provide potentially relevant information about their association. Future reviews may adopt broader eligibility criteria or conduct sensitivity or supplementary analyses including such studies to determine whether they provide convergent or divergent evidence.

Despite these limitations, the present review makes several important contributions to the literature. First, it provides a systematic synthesis of an empirically consistent but conceptually fragmented body of evidence. Second, by jointly considering executive, emotional, and motivational processes, it advances a more integrated framework for understanding the association between ADHD symptoms and procrastination. From this perspective, procrastination may be conceptualized as a transdiagnostic self-regulatory difficulty closely associated with ADHD.

From a clinical perspective, these findings suggest that procrastination may represent a promising intervention target in individuals with elevated ADHD symptoms, consistent with clinical approaches to adult ADHD that have long emphasized procrastination, task initiation difficulties, and difficulties translating intentions into actions as relevant treatment targets (Safren, 2006; Ramsay, 2020). Future intervention research should evaluate whether treatments specifically targeting procrastinatory behavior - through improvements in executive functioning, emotion regulation, motivational processes, or time-management strategies - can produce downstream benefits in everyday functioning and quality of life. Longitudinal and randomized controlled studies will be particularly important for determining whether reducing procrastination contributes to improved functional outcomes in individuals with ADHD.

In conclusion, the available evidence consistently supports the existence of a robust association between ADHD symptoms and procrastination. Although the underlying mechanisms and the directionality of the relationship require further investigation, the findings underscore the importance of considering procrastination within broader models of self-regulation in ADHD and point to relevant directions for future research.

CRediT authorship contribution statement

Rossella Suriano: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Ethics approval

Not applicable.

Funding

No funding was received for conducting this study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ridd.2026.105342.

Data Availability

Data will be made available on request.

References

Altgassen, M., Scheres, A., & Edel, M. A. (2019). Prospective memory (partially) mediates the link between ADHD symptoms and procrastination. *ADHD Attention Deficit and Hyperactivity Disorders, 11*(1), 59–71. https://doi.org/10.1007/s12402-018-0273-x

$1 $2 Association, A.P. A. (2013). Diagnostic and statistical manual of mental disorders.

Ashworth, B., & McCown, W. (2018). Trait procrastination, hoarding, and continuous performance attention scores. *Current Psychology, 37*(2), 454–459. https://doi.org/10.1007/s12144-017-9696-3

Barkley, R. A. (1997). Behavioral inhibition, sustained attention, and executive functions: constructing a unifying theory of ADHD. *Psychological Bulletin, 121*(1), 65.

Bodalski, E. A., Flory, K., Canu, W. H., Willcutt, E. G., & Hartung, C. M. (2023). ADHD symptoms and procrastination in college students: The roles of emotion dysregulation and self-esteem. *Journal of Psychopathology and Behavioral Assessment, 45*(1), 48–57. https://doi.org/10.1007/s10862-022-09996-2

Bolden, J., & Fillauer, J. P. (2020). Tomorrow is the busiest day of the week": Executive functions mediate the relation between procrastination and attention problems. *Journal of American College Health, 68*(8), 854–863. https://doi.org/10.1080/07448481.2019.1626399

Choi, J. N., & Moran, S. V. (2009). Why not procrastinate? Development and validation of a new active procrastination scale. *The Journal of Social Psychology, 149*(2), 195–212. https://doi.org/10.3200/SOCP.149.2.195-212

Chun Chu, A. H., & Choi, J. N. (2005). Rethinking procrastination: Positive effects of "active" procrastination behavior on attitudes and performance. *The Journal of Social Psychology, 145*(3), 245–264. https://doi.org/10.3200/SOCP.145.3.245-264
Fabio, R. A., & Suriano, R. (2021). The influence of media exposure on anxiety and working memory during lockdown period in Italy. *International Journal of Environmental Research and Public Health, 18*(17), 9279.
Fabio, R. A., & Suriano, R. (2024). The role of smartphone use in sensory processing: Differences between adolescents with ADHD and typical development. *International Journal of Environmental Research and Public Health, 21*(12), 1705.
Fabio, R. A., Orsino, C., Lecciso, F., Levante, A., & Suriano, R. (2024). Atypical sensory processing in adolescents with Attention Deficit Hyperactivity Disorder: A comparative study. *Research in Developmental Disabilities, 146*, Article 104674.
Fabio, R. A., Picciotto, G., & Suriano, R. (2026). The role of automatization in complex inductive reasoning: evidence from three experiments. *Psychological Research, 90*(1), 31.
Faraone, S. V., Banaschewski, T., Coghill, D., Zheng, Y., Biederman, J., Bellgrove, M. A., … Wang, Y. (2021). The world federation of ADHD international consensus statement: 208 evidence-based conclusions about the disorder. *Neuroscience & Biobehavioral Reviews, 128*, 789–818. https://doi.org/10.1016/j.neubiorev.2021.01.022
Ferrari, J. R. (1992). Psychometric validation of two procrastination inventories for adults: Arousal and avoidance measures. *Journal of Psychopathology and Behavioral Assessment, 14*(2), 97–110. https://doi.org/10.1007/BF00965170
French, B., Nalbant, G., Wright, H., Sayal, K., Daley, D., Groom, M. J., … Hall, C. L. (2024). The impacts associated with having ADHD: an umbrella review. *Frontiers in Psychiatry, 15*, Article 1343314. https://doi.org/10.3389/fpsyt.2024.1343314
Hofmann, W., Friese, M., & Strack, F. (2009). Impulse and self-control from a dual-systems perspective. *Perspectives on psychological Science, 4*(2), 162–176. https://doi.org/10.1111/j.1745-6924.2009.01116.x
Karalunas, S. L., & Huang-Pollock, C. L. (2011). Examining relationships between executive functioning and delay aversion in attention deficit hyperactivity disorder. *Journal of Clinical Child & Adolescent Psychology, 40*(6), 837–847. https://doi.org/10.1080/15374416.2011.614578
Klingsieck, K. B. (2013a). Procrastination in different life-domains: is procrastination domain specific? *Current Psychology, 32*(2), 175–185. https://doi.org/10.1007/s12144-013-9171-8
Klingsieck, K. B. (2013b). Procrastination. *European psychologist*. https://doi.org/10.1027/1016-9040/a000138
Kosheleff, A. R., Mason, O., Jain, R., Koch, J., & Rubin, J. (2023). Functional impairments associated with ADHD in adulthood and the impact of pharmacological treatment. *Journal of Attention Disorders, 27*(7), 669–697. https://doi.org/10.1177/10870547231158572
Martinussen, R., Hayden, J., Hogg-Johnson, S., & Tannock, R. (2005). A meta-analysis of working memory impairments in children with attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child & Adolescent Psychiatry, 44*(4), 377–384. https://doi.org/10.1097/01.chi.0000153228.72591.73
Marx, I., Hacker, T., Yu, X., Cortese, S., & Sonuga-Barke, E. (2021). ADHD and the choice of small immediate over larger delayed rewards: a comparative meta-analysis of performance on simple choice-delay and temporal discounting paradigms. *Journal of Attention Disorders, 25*(2), 171–187.
Massoodi, A., Kheirkhah, F., Moeini, B., & Ahangar, H. G. (2025). *The Correlations Between Academic Procrastination and Academic Burnout with Symptoms of Adult Attention Deficit Hyperactivity Disorder (ADHD) in Clinical Medical Students in Northern Iran* (No. In Press, p. e165648). *Brieflands*. https://doi.org/10.5812/semj-159756
Mohammadi Bytamar, J., Saed, O., & Khakpoor, S. (2020). Emotion regulation difficulties and academic procrastination. *Frontiers in Psychology, 11*, Article 524588. https://doi.org/10.3389/fpsyg.2020.524588
Müller, V., Mellor, D., & Piko, B. F. (2023). How to procrastinate productively with ADHD: A study of smartphone use, depression, and other academic variables among university students with ADHD symptoms. *Journal of Attention Disorders, 27*(9), 951–959. https://doi.org/10.1177/10870547231171724
Netzer Turgeman, R., & Pollak, Y. (2023). Using the temporal motivation theory to explain the relation between ADHD and procrastination. *Australian Psychologist, 58*(6), 448–456. https://doi.org/10.1080/00050067.2023.2218540
Netzer Turgeman, R., & Pollak, Y. (2025). Adult ADHD-Related Poor Quality of Life: Investigating the Role of Procrastination. *Scandinavian Journal of Psychology, 66*(5), 729–737. https://doi.org/10.1111/sjop.13117
Niermann, H. C., & Scheres, A. (2014). The relation between procrastination and symptoms of attention-deficit hyperactivity disorder (ADHD) in undergraduate students. *International Journal of Methods in Psychiatric Research, 23*(4), 411–421. https://doi.org/10.1002/mpr.1440
Oguchi, M., Takahashi, T., Nitta, Y., & Kumano, H. (2021). The moderating effect of attention-deficit hyperactivity disorder symptoms on the relationship between procrastination and internalizing symptoms in the general adult population. *Frontiers in Psychology, 12*, Article 708579. https://doi.org/10.3389/fpsyg.2021.708579
Petrovic, P., & Castellanos, F. X. (2016). Top-down dysregulation—from ADHD to emotional instability. *Frontiers in Behavioral Neuroscience, 10*, 70. https://doi.org/10.3389/fnbeh.2016.00070
Ramsay, J. R. (2020). *Rethinking adult ADHD: Helping clients turn intentions into actions*. American Psychological Association.
Rebetez, M. M. L., Rochat, L., & Van der Linden, M. (2015). Cognitive, emotional, and motivational factors related to procrastination: A cluster analytic approach. *Personality and Individual Differences, 76*, 1–6. https://doi.org/10.1016/j.paid.2014.11.044
Rebetez, M. M. L., Rochat, L., Barsics, C., & Van der Linden, M. (2018). Procrastination as a self-regulation failure: The role of impulsivity and intrusive thoughts. *Psychological Reports, 121*(1), 26–41. https://doi.org/10.1177/0033294117720695
Safren, S. A. (2006). Cognitive-behavioral approaches to ADHD treatment in adulthood. *Journal of Clinical Psychiatry, 67*, 46.
Shirdel, S., Shadbafi, M., Shirdel, S., & Zarean, M. (2024). Structural relationships of the Adult Attention-Deficit/Hyperactivity symptoms, Sluggish Cognitive Tempo, and driving behavior: mediating role of procrastination. *Current Psychology, 43*(19), 17879–17888. https://doi.org/10.1007/s12144-023-05215-7
Sirois, F. M. (2014). Absorbed in the moment? An investigation of procrastination, absorption and cognitive failures. *Personality and Individual differences, 71*, 30–34. https://doi.org/10.1016/j.paid.2014.07.016
Sirois, F., & Pychyl, T. (2013). Procrastination and the priority of short-term mood regulation: Consequences for future self. *Social and personality Psychology Compass, 7*(2), 115–127. https://doi.org/10.1111/spc3.12011
Sonuga-Barke, E. J. (2002). Psychological heterogeneity in AD/HD—a dual pathway model of behaviour and cognition. *Behavioural brain Research, 130*(1-2), 29–36. https://doi.org/10.1016/S0166-4328(01)00432-6
Sonuga-Barke, E. J. (2003). The dual pathway model of AD/HD: an elaboration of neuro-developmental characteristics. *Neuroscience & Biobehavioral Reviews, 27*(7), 593–604. https://doi.org/10.1016/j.neubiorev.2003.08.005
Steel, P. (2007). The nature of procrastination: a meta-analytic and theoretical review of quintessential self-regulatory failure. *Psychological Bulletin, 133*(1), 65.
Steel, P., & König, C. J. (2006). Integrating theories of motivation. *Academy of Management Review, 31*(4), 889–913. https://doi.org/10.5465/amr.2006.22527462
Steel, P., Svartdal, F., Thundiyil, T., & Brothen, T. (2018). Examining procrastination across multiple goal stages: a longitudinal study of temporal motivation theory. *Frontiers in Psychology, 9*, 327. https://doi.org/10.3389/fpsyg.2018.00327
Svartdal, F., & Nemtcan, E. (2022). Past negative consequences of unnecessary delay as a marker of procrastination. *Frontiers in Psychology, 13*, Article 787337. https://doi.org/10.3389/fpsyg.2022.787337
Netzer Turgeman, R., & Pollak, Y. (2026). What Are You Waiting For?! Roles of Motivation, Goal Orientation, and Emotion Regulation in Explaining the Link Between ADHD and Procrastination. *Journal of Attention Disorders, 30*(5), 694–704.
Willcutt, E. G., Doyle, A. E., Nigg, J. T., Faraone, S. V., & Pennington, B. F. (2005). Validity of the executive function theory of attention-deficit/hyperactivity disorder: a meta-analytic review. *Biological Psychiatry, 57*(11), 1336–1346. https://doi.org/10.1016/j.biopsych.2005.02.006
Wu, H., Gui, D., Lin, W., Gu, R., Zhu, X., & Liu, X. (2016). The procrastinators want it now: Behavioral and event-related potential evidence of the procrastination of intertemporal choices. *Brain and Cognition, 107*, 16–23. https://doi.org/10.1016/j.bandc.2016.06.005
Zentall, T. R. (2021). Basic behavioral processes involved in procrastination. *Frontiers in Psychology, 12*, Article 769928. https://doi.org/10.3389/fpsyg.2021.769928
Zhang, S., Liu, P., & Feng, T. (2019). To do it now or later: The cognitive mechanisms and neural substrates underlying procrastination. *Wiley Interdisciplinary Reviews: Cognitive Science, 10*(4), Article e1492.